CSCI U550 AI and Cybersecurity | Fall 2026 | Dr. Qingli Zeng

Project 1 Datasheet: final_curated_dataset.csv

Template based on Appendix A of the project guide (adapted from Gebru et al., 2021)

Team members	Track (A or B)
Ana Madrigal, Sydney Dailey, Katherine Vykhovanets	Track A: Phishing URLs
Collection window (first to last pull)	Final size: 51072 phishing / 51072 legitimate

1. Motivation

The dataset was created to train a model to detect phishing URLs by being able to distinguish between legitimate and phishing URLs. Based on Mitchell's P, T, and D framework, the task is to classify URLs as legitimate or phishing. The data/experience is the final curated and labeled dataset. The performance is measured using precision, recall, F1, ROC-AUC, PR-AUC, and recall at 1% and 0.01% false positive rates.

2. Composition

Each instance represents a collected URL. There are 51,072 instances in both the legitimate and the phishing class. This is a sample of the larger population of real-world URLs since they came from an Open Phish collection repo, and they may not represent every possible type of phishing. The legitimate URLs came from a pre-collected CSV. No exact duplicates or near duplicates are left in the final curated dataset.

3. Collection process

The provided collect_phish.py script was used to collect the phishing URLs from the OpenPhish feed and its public history to create the phish_urls.csv. The collection_log.csv records the history of backfill and live pulls with the dates and number of new URLs. The live pulls were pulled on five different days since the feed only updates twice a day and we did not interact with or visit the URLs. The provided collection of benign URLs (benign_urls.csv) was used as the project's legitimate URLs. After preprocessing and deduplication, we combined both sources into a final dataset, labeling each entry as legitimate or phishing based on the source CSV.

4. Preprocessing and cleaning

The original datasets were preserved, and separate Python scripts were used to clean the URLs without changing the raw files. We cleaned the trailing slashes, query order, tracking parameters, hostname capitalization, percent encoding, fragments, and removed the scheme from the phishing URLs used for modeling. After removing near-duplicates, we sampled the large benign dataset so both classes had the same number of URLs and the same path-depth distribution.

Step	Rule applied	Removed	Remaining
Exact duplicate URLs / messages	Code that fetches logs eliminates all exact duplicates	0 phishing, 0 legitimate	51,209 phishing, 344,800 legitimate
Near-duplicates (trailing slash, query order, tracking parameters)	After undergoing normalization of trailing slash, query order, tracking parameters, lowercase host, decode % encoding and strip fragments, URLs were labeled as near-duplicates if they became identical	137 phishing, 8 legitimate	51,072 phishing, 344,792 legitimate
Normalization (lowercase host, decode %-encoding, strip fragments)	Normalize trailing slash/ query order, remove tracking parameters and fragments, lowercase host, decode % encoding, and strip phishing scheme from model representation	0 phishing, 0 legitimate	51,072 phishing, 344,792 legitimate
Class balancing / subsampling decision	Keep all 51,072 phishing URLs. Sample legitimate URLs with seed 42 to match phishing class size and path-depth buckets	293,720 legitimate	51,072 phishing, 51,072 legitimate

5. Leakage and shortcuts

We found a few differences that could make the model recognize which dataset came from instead of learning actual phishing patterns. For example, the original phishing URLs all had a scheme, while none of the original

legitimate URLs had schemes. Both classes also had very different path depths. So, we removed the scheme from the phishing URLs and sampled the benign URLs to match the phishing path-depth distribution. The differences were kept in TLDs and domain concentration because they could be related to real phishing behavior, but we documented them as possible shortcuts.

Top 20 domains per class (by number of samples)

#	Phishing domain	Count	Legitimate domain	Count
1	pages.dev	7113	blogspot.com	2494
2	vercel.app	4302	youtube.com	1249
3	github.io	3399	facebook.com	989
4	blogspot.com	2806	yahoo.com	482
5	icefactory.cl	959	wikipedia.org	463
6	weebly.com	921	amazon.com	452
7	netlify.app	920	mylife.com	425
8	roblox.com.ml	655	myspace.com	410
9	workers.dev	586	wordpress.com	409
10	webflow.io	543	wn.com	279
11	framer.app	429	manta.com	172
12	replit.app	388	linkedin.com	171
13	roblox.com.bi	366	imdb.com	168
14	godaddysites.com	355	famouswhy.com	157
15	roblox.com.mu	318	spokeo.com	153
16	backblazeb2.com	317	thefreedictionary.com	149
17	typedream.app	312	twitter.com	140
18	roblox.com.bn	302	tripod.com	139
19	edgeone.dev	271	ancestry.com	139
20	robiox.com.py	271	foodpages.ca	137

Class comparability check

Property	Phishing set	Legitimate set	Signal or artifact?	Action taken
Scheme present (http/https)	Original: 100% Model representation: 0%	0%	Collection artifact	Stripped the schemes from the phishing model so both classes have 0 scheme presence
Path present / path depth	Present: 45.34% Depth : 0	Present: 45.35% Depth: 0	Collection artifact in the original files	Stratified benign subsampling by path depth bucket to match phishing distribution
TLD mix	.com – 23.36% .dev – 16% .app – 13.68%	.com – 76.37% .org – 8.65% .net – 4.05%	Mixed real signal / source artifact	Preserved and documented instead of forcing distributions to match
Age of the data	Current/live collection	Older supplied corpus	Collection artifact/ limitation	Documented

Property	Phishing set	Legitimate set	Signal or artifact?	Action taken
Domain concentration	Top 20: 49.99%	Top 20: 17.97%	Potential shortcut and real hosting concentration	Preserved and documented; domain-grouped evaluation used to test generalization

6. Uses and limitations

This dataset is great for training and testing models that are used to classify URLs as legitimate or phishing based on the URL itself. However, it should not be used as a complete representation of all phishing, and it should not be assumed to perform the same on future URLs, especially since the legitimate data is older than the phishing data. The dataset can become outdated quickly because phishing URLs and patterns change constantly.

7. Distribution and ethics

The dataset contains URLs, but it does not contain personal information. The full dataset should not be shared or uploaded to the public. Only the 200-row sample will be shared in the project ZIP folder. The full dataset is available on each teammate's OneDrive.